PAPER 6A · 50 MARKS

The Treasury Desk

Is this worth funding, and with whose money?

Financial Management. Ratios, sources of finance and cost of capital, capital structure and leverage, investment appraisal and dividend policy, and the working capital cycle.

4	4	12	11,262
SECTIONS	ATTEMPTS	SEALED ANSWERS	WORDS

Unit 1 — Financial Management and Financial Analysis

PRINCIPAL

> Profit is up 18%. The bank has refused to renew the working capital limit. Explain that.

YOU

> If profit is up, the bank is being unreasonable.

PRINCIPAL

> The bank lends against cash and the balance sheet, not against profit. Find where the profit went.

MARGIN NOTE

> *Unit 1 in one exchange. Profit maximisation and wealth maximisation are not the same objective, and ratios say which one the entity is achieving.*

The finance function and its objective

Financial management plans, raises, controls and administers funds. Its scope is **three decisions and one constraint**: the **financing decision** (how much, from where, in what mix — Unit 2), the **investment** and **dividend** decisions (Unit 3), and the **liquidity decision** (Unit 4).

	PROFIT MAXIMISATION	WEALTH MAXIMISATION
Measure	Accounting profit or EPS	Market value of the owners' stake
Timing	Indifferent to when profit arrives	Discounts for timing

	PROFIT MAXIMISATION	WEALTH MAXIMISATION
Risk	Ignored	Priced in the discount rate
Basis	Accrual profit, an accounting figure	Cash flow, an economic figure
Vagueness	Which profit — pre-tax or post, short or long run?	One number, quoted daily
Verdict	A yardstick, not an objective	The accepted objective

The three objections to profit maximisation compress to one phrase: **timing, risk and vagueness.**

Value creation is wealth maximisation made operational — value is created when return on capital employed exceeds the cost of that capital. Unit 2 measures that cost, Unit 3 tests projects against it, Unit 4 asks the same of money tied up in stock and debtors.

REMEMBER THIS

Value is created only when return on capital employed exceeds the cost of capital. Growth financed above its own return destroys wealth while still raising profit.

> Funds ploughed into projects earning 9% while capital costs 12% raise profit and lower market value. That asymmetry is the mark when a question contrasts the two objectives.

Financial manager against financial controller, examined as a contrast:

	FINANCIAL MANAGER	FINANCIAL CONTROLLER
Orientation	Forward-looking, decision-taking	Backward-looking, reporting
Faces	Banks, investors, the capital market	Management, auditors, tax authorities
Does	Raises funds, sets structure, appraises investment, manages cash and credit	Records, budgets, prepares statements, runs internal control, reports variances

Functions of finance executives: estimating fund requirements; deciding capital structure; selecting sources; investment of funds; maintaining liquidity; disposal of surplus and the dividend decision; financial negotiation; evaluating performance; and tracking the company's own share price.

The **financial management environment** is where those decisions are taken: the **money market** for short-term instruments, the **capital market** for long-term funds, the intermediaries channelling savings into investment, and the regulatory, tax and interest-rate framework. The firm is largely a **price-taker** here — the cost of capital in Unit 2 is set in these markets, not chosen.

Financial distress and insolvency

Financial distress is the condition in which cash flows are insufficient to meet obligations. It is a matter of degree, running from mild liquidity strain through stretched creditors and breached covenants to insolvency. **Insolvency** takes two forms and questions turn on the distinction: **cash-flow (technical) insolvency**, where debts cannot be paid as they fall due even though assets exceed liabilities; and **balance-sheet insolvency**, where liabilities exceed the realisable value of assets.

The **costs of distress** split the way marks do. **Direct** — legal, administrative and professional fees, and management time spent on creditors. **Indirect** — customers and suppliers withdraw, credit tightens, staff leave, assets are sold below value, new finance costs more. The indirect usually exceed the direct.

> ***Verify:*** *Section A treats distress and insolvency as a financial condition and its costs. The statutory resolution machinery sits elsewhere — check the Study Guidelines for your attempt before adding legal detail to an FM answer.*

Financial analysis through ratios

Users, each paired with their question: management (control and decisions), investors (return and risk), lenders (security and servicing capacity), suppliers (short-term solvency), employees (stability), government (compliance and tax), analysts and rating agencies.

Sources of financial data: the balance sheet and statement of profit and loss, the cash flow statement, notes and accounting policies, the directors' report and management discussion; and internally, budgets, cost records and management reports, plus industry data to compare against.

FAMILY	RATIOS	DIAGNOSES
Liquidity	Current, quick, cash	Can short-term obligations be met

FAMILY	RATIOS	DIAGNOSES
Solvency / leverage	Debt–equity, debt to total assets, proprietary, capital gearing, interest coverage, DSCR	How much risk sits with lenders, and can they be serviced
Activity / turnover	Inventory, receivables, payables, fixed asset, working capital, total asset	How hard the assets are worked
Profitability	Gross, operating, net profit; ROCE, ROE, return on assets	Whether trading and capital both pay
Market / valuation	EPS, price–earnings, earnings yield, dividend yield, payout	What the market pays for the earnings

FORMULA

Return on Capital Employed

MEASURES

The return earned on all long-term funds in the business, before deciding who gets it.

FORMULA

ROCE = (EBIT / Capital employed) x 100
Capital employed = Shareholders' funds + Long-term debt = Total assets - Current liabilities

TERMS

EBIT = earnings before interest and tax. Shareholders' funds = equity + preference capital + reserves, less accumulated losses.

WATCH OUT

The numerator is before interest, because interest is a return to one of the suppliers of the capital in the denominator. PAT over capital employed mixes a post-interest return with a pre-interest base — the commonest error here.

FORMULA

Return on equity, decomposed (DuPont)

MEASURES

Splits the owners' return into a trading, an efficiency and a gearing effect.

FORMULA

ROE = PAT / Shareholders' funds = (PAT / Sales) x (Sales / Total assets) x (Total assets / Shareholders' funds) = Net profit margin x Total asset turnover x Equity multiplier

TERMS

Where preference shares exist, return on *equity* funds uses PAT less preference dividend over equity shareholders' funds only.

WATCH OUT

The three factors must multiply back to the direct ROE; if not, you have mixed average and closing bases. An ROE rising only through the third factor is a gearing story, not an improvement.

FORMULA

Interest coverage and debt service coverage

FORMULA

Interest coverage = EBIT / Interest DSCR = (PAT + Depreciation + Interest on term loan) /

(Interest on term loan + Principal instalment)

WATCH OUT

Both are "times", never percentages. DSCR adds principal repayment, which is not an expense and so is absent from profit, and adds back non-cash depreciation. Comfortable interest coverage with a DSCR below one is the company in the dialogue.

WORKED PROBLEM

A complete ratio pass over one small set of accounts

GIVEN

All figures ₹ lakh. Equity share capital (face ₹10) 200; reserves 100; 12% debentures 200; trade payables 90; other current liabilities 10. Net fixed assets 350; inventory 100; receivables 80; cash 70. Sales (all credit) 1,000; cost of goods sold 700; operating expenses 150; purchases

1. Tax 30%. Use closing figures throughout.

ATTEMPT IT

Compute the profit line to PAT, then current and quick ratios, debt–equity, interest coverage, inventory and debtor velocity in days, creditors' days, ROCE, net profit ratio, ROE and EPS. Reconcile ROE through the DuPont decomposition.

THE SOLUTION

Compute the profit line to PAT, then current and quick ratios, debt–equity, interest coverage, inventory and debtor velocity in days, creditors' days, ROCE, net profit ratio, ROE and EPS. Reconcile ROE through the DuPont decomposition.

SOLUTION

Balance the sheet first: shareholders' funds ₹300; capital employed 300 + 200 = ₹500; current liabilities ₹100; current assets ₹250; total 600 = 350 + 250. It balances.

Gross profit = 1,000 − 700 = ₹300. EBIT = 300 − 150 = ₹150. Interest = 12% x 200 = ₹24. PBT = ₹126. Tax at 30% = ₹37.80. PAT = ₹88.20.

Current ratio = 250 / 100 = **2.5 : 1**; quick = (250 − 100) / 100 = **1.5 : 1**.

Debt–equity = 200 / 300 = **0.67 : 1**; interest coverage = 150 / 24 = **6.25 times**.

Inventory turnover = 700 / 100 = 7 times → **52 days** (365 / 7).

Debtor turnover = 1,000 / 80 = 12.5 times → **29 days**. Creditors = 90 / 720 x 365 = **46 days**.

ROCE = 150 / 500 = **30%**; net profit ratio = 88.20 / 1,000 = **8.82%**.

ROE = 88.20 / 300 = **29.4%**. Shares = 20 lakh, so EPS = **₹4.41**.

DuPont: 8.82% x (1,000 / 600 = 1.6667) x (600 / 300 = 2.0) = 0.0882 x 1.6667 x 2 = 0.2940 = **29.4%**, equal to the direct ROE.

CHECK

Three checks. The DuPont product must return the direct ROE to the decimal — it does. ROCE (30%) exceeds the pre-tax cost of debt (12%), so gearing works for the owners, consistent with ROE landing close to ROCE despite tax. And collection (29 days) is faster than payment (46 days), so suppliers finance

the firm — if yours reversed, check which turnover you divided by which.

EXAM TRAP

Mixing closing balances with averages inside one answer

Turnover ratios are a flow over an average stock, so averages are preferable and questions often say "average inventory". Given only a closing balance sheet, closing figures are the only honest basis — state the assumption once and hold it across every ratio. The mark is for the disclosure and the consistency. Same for 365 against 360 days: pick one, say so, never switch.

QUICK RECALL

Debtor velocity is 29 days but the stated credit policy is 60 days. Is the ratio wrong?

No, and the finding is favourable. Always compare a ratio against **policy, prior year and industry** —

alone it is a number, not a diagnosis. The reverse case is the examinable one: collection beyond policy points to weak credit control and possibly unrecognised bad debts.

Limitations of ratio analysis is a stock question: ratios are historical and say nothing about the future; no single correct standard exists to compare against; accounting policy differences on depreciation and inventory destroy comparability; price-level changes distort trends; window dressing flatters liquidity; a balance sheet ratio is a single-date snapshot hiding seasonality; aggregation conceals the mix inside a total; qualitative factors are invisible; and a ratio shows a **symptom**, never a cause.

PRACTISE IT

Net profit ratio has fallen from 10% to 8% while ROE has risen from 24% to 27%. Explain how both can be true, and state which ratios you would compute next.

MODEL ANSWER

Compare against what a full-mark answer contains

Write the identity first — **ROE = net profit margin x total asset turnover x equity multiplier** — and the rest is mechanical: with margin down, ROE still rises if assets turn faster, or if the equity multiplier has risen through more debt or a return of capital. Then name the next computations:

- **Total asset turnover** and the individual turnover ratios test the efficiency story; a genuine improvement shows as higher inventory and receivable turnover.
- **Debt–equity, interest coverage and DSCR** test the gearing story. If the multiplier did the work, ROE rose because risk rose.
- **ROCE** is the referee, being unaffected by financing mix. Rising ROCE supports efficiency; ROCE flat while ROE rises is gearing.

Close with the sentence examiners look for: an ROE gain achieved purely through leverage is not value creation, because the equity holder's required return rises with gearing.

Verify: *which ratios are examinable, and whether averages or closing balances are expected, are settled by the Study Material and Study Guidelines for your attempt. These notes revise; they do not replace either.*

Unit 2 — Financing Decisions and Cost of Capital

YOU

> Shares are cheaper than the bank's 12% loan. They cost nothing — we pay a dividend only if we want to.

PRINCIPAL

> Equity is the most expensive money a company can raise. Tell me why by Friday.

MARGIN NOTE

> *The equity holder is paid last and so demands the most, and interest is deductible while dividend is not. That is the spine of the unit.*

Sources of finance

	EQUITY SHARES	PREFERENCE SHARES	DEBT
Return	Residual, discretionary	Fixed rate, ranks before equity	Fixed, contractual

	EQUITY SHARES	PREFERENCE SHARES	DEBT
Maturity	Perpetual; no default possible	Redeemable or irredeemable; arrears accumulate	Dated; enforceable
Tax	Dividend is a post-tax appropriation	Same	Interest is a deductible charge
Control	Votes, so dilutes control	Limited votes	No votes, but covenants bind
Cost to firm	Highest	Between	Lowest

Methods of raising long-term finance: public issue, rights issue, private placement and preferential allotment, offer for sale, debentures or bonds, term loans and deferred credit, and internal accruals.

Short-term sources: trade credit and accrued expenses (spontaneous, and free unless a discount is forgone), bank finance (cash credit, overdraft, bills discounting, working capital demand loan, letter of credit), commercial paper, certificates of deposit, inter-corporate deposits, public deposits, factoring, and advances from customers.

Contemporary sources of funding are examined descriptively, so carry one clause each. **P2P lending** — a platform matches individual lenders directly with

borrowers, cutting out the bank. **Crowd funding** — many small contributors fund a venture through a platform, in donation, reward, debt or equity form. **Equity funding** — a stake issued to angel investors, private equity or strategic investors. **Start-up funding** — the staged ladder from bootstrapping and family money through incubators, accelerators and angel rounds to venture capital, with government schemes alongside.

Internal funds are retained earnings plus funds released by depreciation: no issue cost, no dilution — but not free, because retained earnings carry an opportunity cost equal to what shareholders could have earned, so their cost is the cost of equity.

International sources: American and Global Depository Receipts, Foreign Currency Convertible Bonds, External Commercial Borrowings, foreign currency term loans, Euro issues, export credit, and foreign direct or portfolio investment.

Other sources. Lease financing gives use without ownership; a **finance lease** transfers substantially all risks and rewards, an **operating lease** does not. **Sale and lease back** — the owner sells an asset and immediately leases it back, freeing locked-up capital while keeping the use of it. **Convertible debt** — debt carrying a right to convert into equity: cheap while debt, dilutive once

converted. **Venture capital** — equity or quasi-equity in unlisted high-risk ventures, with active involvement and a planned exit. **Grants** — non-repayable, usually purpose-restricted.

Cost of capital

The cost of capital is the minimum return a project must earn to leave shareholders no worse off. It is the **discount rate in Unit 3**, the **benchmark for value creation** in Unit 1, and the quantity minimised in capital structure design. **Factors**: the general level of interest rates, the tax regime, capital structure and financial policy, business and financial risk, dividend policy, and issue costs.

Say this once and never break it. In WACC, **debt is taken after tax**, because interest is deductible in computing taxable profit, so the firm parts with only `I(1 - t)`. **Equity and preference are not**, because dividends are an appropriation of post-tax profit and carry no shield. Multiplying Ke or Kp by (1 − t) is a guaranteed lost mark.

FORMULA

Cost of equity — dividend growth model and CAPM

FORMULA

Ke = D1/P0 + g with issue cost: Ke = D1/NP + g D1 = D0 x (1 + g) Ke = Rf + Beta x (Rm - Rf)

TERMS

g = constant dividend growth = retention ratio (b) x return on equity (r).

WATCH OUT

D1, not D0, goes in the numerator. If the risk **premium** is given, do not subtract Rf from it again — the commonest CAPM slip. CAPM needs no dividend and no growth, so use it for a non-payer.

FORMULA

Cost of debt and of preference shares

FORMULA

Kd irredeemable = I(1 - t)/NP (drop the (1 - t) for the pre-tax cost) Kd redeemable = [I(1 - t) + (RV - NP)/n] / [(RV + NP)/2] Kp irredeemable = PD/NP Kp redeemable = [PD + (RV - NP)/n] / [(RV + NP)/2]

TERMS

NP = net proceeds after discount and expenses; RV = redemption value; n = years to redemption.

WATCH OUT

Interest is computed on **face value** but divided by **net proceeds**. No (1 − t) appears anywhere in Kp. The redeemable formulas spread the capital gain or loss over the life and over average funds — an *approximation* to the true after-tax IRR, so say which route you used.

FORMULA

Weighted average and marginal cost of capital

FORMULA

WACC (Ko) = (We x Ke) + (Wp x Kp) + (Wd x Kd after tax) + (Wr x Kr) Marginal cost of capital = weighted cost of the ADDITIONAL funds, at new costs and new weights only.

WATCH OUT

Weights (book or market) must total exactly 1.0 and every cost must be on the same tax

basis. Market weights are preferred because each component cost is itself derived from market prices. Marginal cost is not the new average cost of the enlarged pool — a different figure the question may also want.

WORKED PROBLEM

WACC on book-value and market-value weights, then the marginal cost

GIVEN

Capital (₹ lakh) — equity capital 100, retained earnings 50, 10% preference 50 (net proceeds ₹95 per ₹100), 12% irredeemable debentures 100 (net proceeds ₹96 per ₹100). Tax 30%. Equity dividend just paid ₹4, growing at 5%; market price ₹50. Market values: equity 250, preference 45, debentures 105. The company now raises ₹100 lakh as 50% equity (costing 15% after expenses), 20% preference at 11%, 30% debt at 9.5% after tax.

ATTEMPT IT

Compute Ke, Kp and Kd; WACC on book and on market weights; and the marginal cost of

capital. Say which figure appraises a new project.

THE SOLUTION

Compute Ke, Kp and Kd; WACC on book and on market weights; and the marginal cost of capital. Say which figure appraises a new project.

SOLUTION

D1 = 4 x 1.05 = ₹4.20, so **Ke = 4.20/50 + 0.05 = 13.40%**; retained earnings take the same 13.40%, there being no issue cost to deduct. **Kp = 10/95 = 10.53%**, no tax adjustment. **Kd = 12(1 − 0.30)/96 = 8.75%** after tax, against 12/96 = 12.50% before tax.

	BOOK VALUES				MARKET VALUES			
	₹ lakh	Weight	Cost	Product	₹ lakh	Weight	Cost	Product
(bracketed book figures are components of the 150, not additions to the 300)								
Equity capital	(100)				250	0.6250	13.40%	8.375
Retained earnings	(50)				-	-	-	-
Equity + reserves	150	0.5000	13.40%	6.700				
Preference shares	50	0.1667	10.53%	1.754	45	0.1125	10.53%	1.184
12% Debentures	100	0.3333	8.75%	2.917	105	0.2625	8.75%	2.297
	300	1.0000		11.37%	400	1.0000		11.86%

Marginal cost of capital = (0.50 x 15%) + (0.20 x 11%) + (0.30 x 9.5%) = 7.50 + 2.20 + 2.85 = **12.55%**. Appraise the new project at **12.55%**,

because that is the cost of the money the project will actually consume; the historic WACC prices funds already raised.

CHECK

The market WACC exceeds the book WACC only because equity — the dearest source — carries a bigger weight at ₹250 lakh than at ₹150 lakh; if yours came out lower, you have added reserves into the market column too. And each weight column must total exactly 1.0000.

EXAM TRAP

Giving retained earnings a market value of their own

Retained earnings have a book value but no market quotation — the share price already capitalises them, which is why the market column shows a dash. Some Study Material questions instead ask you to **split** the market value of equity between capital and reserves in the ratio of their book values; that convention exists and yields the same WACC,

> both slices carrying the same cost. State which you followed, never both.

> ***Verify:*** *whether any tax on dividend distribution is loaded into Kp or Ke, and whether retained earnings are discounted for personal tax and brokerage, are settled by the Study Material for your attempt. Do not assume a rate.*

Capital structure decisions

Capital structure is the mix of long-term debt and equity. It moves both WACC and the risk borne by equity; the **optimum capital structure** is the mix at which WACC is lowest and firm value highest.

Determinants: cost of capital and the tax shield; business risk and stability of earnings; the need to retain control; flexibility and unused debt capacity; lenders' covenants; size and stage of the company; asset structure and available security; market conditions; and the purpose of the finance. Designing an optimum structure means maximising EPS or value while keeping fixed charges serviceable — carry **FRICT**: flexibility, risk, income, control, timing.

THEORY	KE AND KD AS DEBT RISES	WACC	VALUE OF FIRM
Net Income	Both constant	Falls	Rises with debt; structure is relevant
Net Operating Income	Ke rises just enough to offset; Kd constant	Constant	Unaffected; structure is irrelevant
Traditional	Ke rises slowly then sharply; Kd rises after a point	U-shaped	Optimum at the bottom of the U
Modigliani–Miller	Ke rises per the arbitrage argument; Kd constant	Constant without tax, falls with tax	Vu without tax; Vu + Dt with corporate tax

The NOI mechanism deserves one arithmetic line, because it *is* the argument. With EBIT ₹10,00,000 and Ko 12.5%, value = 10,00,000/0.125 = ₹80,00,000. Add ₹20,00,000 of 10% debt: equity is worth ₹60,00,000, so Ke = (10,00,000 − 2,00,000)/60,00,000 = 13.33%. Recheck the WACC — 0.25 x 10% + 0.75 x 13.33% = **12.5%**, unchanged. Cheap debt bought nothing because equity repriced. MM defends the result by **arbitrage**: an investor can create or undo gearing personally, so two identical firms cannot sell at different prices.

EBIT–EPS analysis chooses between financing plans by comparing EPS at the expected EBIT. The **indifference (break-even EBIT) point** is the EBIT at which two plans give equal EPS. The **financial break-even EBIT** is a

different thing — the EBIT at which EPS is exactly zero, equal to interest plus preference dividend grossed up by (1 − t).

REMEMBER THIS

Above the indifference EBIT, gear up; below it, issue equity. Debt magnifies EPS in both directions, so the choice is really a judgement on how firm the EBIT forecast is.

Worked: to raise ₹40,00,000 on 2,00,000 existing shares, Plan A issues 40,000 shares at ₹100 and Plan B issues 10% debt; tax 30%. Setting EBIT(1−t)/2,40,000 = (EBIT − 4,00,000)(1−t)/2,00,000 gives EBIT = **₹24,00,000**, where both yield EPS of exactly **₹7.00**. Plan B's financial break-even EBIT is ₹4,00,000.

Over- and under-capitalisation compare earnings with capital employed, and candidates reverse them. **Over-capitalised**: capital employed exceeds what earnings justify, the earning rate is below normal, book value of shares exceeds real value — caused by over-issue of capital, assets bought at inflated prices, promotion in a boom, or inadequate depreciation; remedied by

reorganising capital, redeeming debt and writing down assets. **Under-capitalised**: earnings are high relative to capital employed, the earning rate is above normal, real value exceeds book value — caused by under-estimating requirements, conservative dividends building reserves, or high efficiency; remedied by bonus shares, share splits or a higher dividend.

Leverages

FORMULA

The three leverages and the identity between them

FORMULA

DOL = Contribution / EBIT = %change in EBIT / %change in Sales DFL = EBIT / PBT = %change in EPS / %change in EBIT DFL with preference shares = EBIT / [EBIT - I - Dp/(1 - t)] DCL = DOL x DFL = %change in EPS / %change in Sales DCL = Contribution / PBT <- - ONLY when there is no preference dividend DCL with preference shares = Contribution / [EBIT - I - Dp/(1 - t)]

TERMS

Contribution = Sales − Variable cost. PBT = EBIT − Interest. Dp = preference dividend.

WATCH OUT

Operating leverage comes from fixed **operating** cost, financial leverage from fixed **financial** charges. A firm with no fixed cost has DOL of 1 and one with no debt has DFL of 1 — leverage is a multiplier, so its neutral value is one, never zero. DCL is a product, never a sum.

The shortcut `Contribution / PBT` is only valid when `Dp = 0`, and this is the trap most revision sheets get wrong. It comes from cancelling EBIT out of `(Contribution/EBIT) x (EBIT/PBT)` — but the moment a preference dividend exists, DFL is no longer `EBIT/PBT`, so there is no EBIT to cancel and the shortcut silently understates DCL. Use `%change in EPS / %change in Sales` as your check; it is the one route that is always right, because DCL is *defined* by the effect on EPS and preference dividend reduces earnings available to equity.

PRACTISE IT

Sales ₹10,00,000, variable cost ₹6,00,000, fixed cost ₹2,00,000, interest ₹50,000, preference dividend ₹21,000, tax 30%, 1,00,000 equity shares. Compute the three leverages by both routes, reconcile them, and verify DFL from EPS.

MODEL ANSWER

Compare against what a full-mark answer contains

Lay the statement out first, because both routes read off it.

	Base	Sales +10%	
Sales	10,00,000	11,00,000	
Less: Variable cost	6,00,000	6,60,000	
CONTRIBUTION	4,00,000	4,40,000	
Less: Fixed cost	2,00,000	2,00,000	
EBIT	2,00,000	2,40,000	(+20%)
Less: Interest	50,000	50,000	
PBT	1,50,000	1,90,000	(+26.67%)
Less: Tax at 30%	45,000	57,000	
PAT	1,05,000	1,33,000	
Less: Preference dividend	21,000	21,000	
Earnings for equity	84,000	1,12,000	
EPS (1,00,000 shares)	0.84	1.12	(+33.33%)

Operating leverage. DOL = 4,00,000/2,00,000 = **2.00**. Check: EBIT is up 20% on a 10% sales rise, so 20/10 = 2 ✓.

Financial leverage — and this is where the preference dividend bites. The dividend is paid out of post-tax profit, so it must be grossed up before it can sit alongside pre-tax interest: 21,000/0.70 = ₹30,000. Then

DFL = 2,00,000 / (2,00,000 − 50,000 − 30,000) = 2,00,000/1,20,000 = **1.6667**

not EBIT/PBT = 2,00,000/1,50,000 = 1.3333. Check it against EPS: EPS rises 33.33% while EBIT rises 20%, and 33.33/20 = **1.6667 ✓**. The grossed-up figure is right and EBIT/PBT is wrong here.

Combined leverage. DCL = DOL x DFL = 2.00 x 1.6667 = **3.3333.** Check against the percentages: EPS is up 33.33% on a 10% sales rise, so 33.33/10 = **3.3333 ✓**.

Now notice what you must *not* do. Contribution/PBT = 4,00,000/1,50,000 = 2.6667, and that is **not** the answer to this question. The shortcut assumes

DFL is EBIT/PBT so that EBIT cancels out of the product — which stops being true the moment a preference dividend exists. The correct direct route here is

DCL = Contribution / [EBIT − I − Dp/(1 − t)] = 4,00,000/1,20,000 = **3.3333 ✓**

Three routes, one answer. **The 26.67% rise in PBT is a red herring**: it gives you nothing, because DCL is defined on EPS, and between PBT and EPS sits the preference dividend.

QUICK RECALL

Two firms have equal EBIT and equal DCL. One has DOL 3 and DFL 1.2, the other DOL 1.2 and DFL 3. Which is riskier?

Sensitivity to sales is identical; risk is not. The second firm's exposure is **financial** — fixed interest payable whatever happens — so a bad year threatens solvency, not just profit.

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UNIT 2 FORMULA SHEET — cost of capital, capital structure, leverage
   I = annual interest in ₹ on face value      t  = tax rate
   NP = net proceeds after discount/expenses   RV = redemption value      n = years to redemption
   PD = annual preference dividend in ₹        Dp = preference dividend in the EPS line

COST OF EACH SOURCE
  Ke  dividend growth model     = D1/P0 + g                D1 = D0 x (1 + g)
  Ke  with floatation cost      = D1/NP + g                NP = P0 - cost per share
  Ke  CAPM                      = Rf + Beta x (Rm - Rf)
  Ke  earnings yield            = EPS / Net proceeds per share
  Kr  retained earnings         = Ke  (less personal tax and brokerage if the question says so)
  Kp  irredeemable              = PD / NP                        -- never x (1 - t)
  Kp  redeemable                = [PD + (RV - NP)/n] / [(RV + NP)/2]
  Kd  irredeemable, after tax   = I(1 - t) / NP
  Kd  irredeemable, before tax  = I / NP
  Kd  redeemable, after tax     = [I(1 - t) + (RV - NP)/n] / [(RV + NP)/2]
  Kd  term loan                 = Interest rate x (1 - t)
  Growth rate g                 = retention ratio (b) x return on equity (r)

COMBINING THEM
  WACC (Ko)                     = Sum of (weight x cost) over all sources; weights total 1.0
  Weights                       = book values OR market values (market preferred)
  Marginal cost of capital      = weighted cost of the ADDITIONAL funds only

CAPITAL STRUCTURE
  Value of firm (NOI / MM, no tax)  V  = EBIT / Ko            Ke = (EBIT - I) / Equity
  MM with corporate tax             Vl = Vu + (Debt x t)
  EPS                               = [ (EBIT - I)(1 - t) - Dp ] / No. of equity shares
  Indifference (break-even) EBIT    [(E-I1)(1-t) - Dp1] / n1 = [(E-I2)(1-t) - Dp2] / n2
  Financial break-even EBIT         = I + Dp/(1 - t)                (the EBIT at which EPS = 0)

LEVERAGE
  DOL = Contribution / EBIT             = %change EBIT / %change Sales
  DFL = EBIT / PBT                      = %change EPS  / %change EBIT
  DFL with preference shares            = EBIT / [EBIT - I - Dp/(1 - t)]
  DCL = DOL x DFL                       = %change EPS  / %change Sales
        = Contribution / PBT            ONLY if there is no preference dividend
        = Contribution / [EBIT - I - Dp/(1-t)]      when there is one
  Neutral value of every leverage is 1.0, not 0. DCL is a product, never a sum.
```

Unit 3 — Capital Investment and Dividend Decisions

Capital investment decisions commit large sums, are largely irreversible, and bind the firm for years, so the **objective** is to accept only projects returning more than the capital funding them costs, and to rank projects when funds are short. Two rules govern everything here: work with **incremental cash flows after tax**, never accounting profit, and ignore sunk costs while including opportunity costs. Depreciation enters only through its **tax saving**, not being a cash outflow.

FORMULA

Payback, discounted payback and Accounting Rate of Return

FORMULA

Payback period = Years fully recovered + (Unrecovered cost / Cash flow of next year) Discounted payback = the same, on cash flows already discounted at the cost of capital ARR = (Average annual accounting profit after tax / Investment) x 100 Average

investment = (Initial outlay + Salvage value) / 2

TERMS

Accounting profit = cash flow − depreciation. With even flows, payback = Outlay / Annual inflow.

WATCH OUT

Payback uses **cash flows**, ARR uses **profit after depreciation** — mixing the two is the commonest error in this unit. Payback ignores everything after the cut-off and ignores the time value of money; discounted payback fixes the second defect, not the first.

Verify: *the ARR denominator is a genuine split — some editions divide by* ***average*** *investment, others by* ***initial*** *investment, and the answers differ by a factor of two. Average investment is more commonly expected in ICAI solutions. State your base, and confirm it for your attempt.*

FORMULA

Net Present Value and Profitability Index

FORMULA

NPV = Sum of [Cash inflow of year t / (1 + k)^t] - Initial outlay PI (Benefit-Cost Ratio) = PV of future cash inflows / Initial outlay

TERMS

k = cost of capital. PI = 1 + (NPV / Initial outlay).

WATCH OUT

Accept if NPV is positive. PI and NPV always agree on accept-or-reject, PI exceeding 1 exactly when NPV exceeds 0 — but they can **disagree on ranking**, NPV measuring rupees created and PI rupees created per rupee invested.

Strengths of NPV: uses all cash flows, discounts them properly, is in absolute rupees and so is additive across projects, and uses the cost of capital directly. **Limitations**: needs an estimate of the cost of capital; is unintuitive to non-finance managers; ignores the size of the outlay (₹10,000 of NPV counts the same on a ₹1 lakh project as on a ₹1 crore one); and assumes intermediate flows are reinvested at the cost of capital.

The working capital adjustment in NPV has three rules: the working capital is an **outflow when committed**, it is **not depreciated**, and it is **recovered as an inflow at the end** of the life, **untaxed**, because releasing stock and debtors is not income.

> ***Verify:*** *conventions vary. Some questions inject the whole amount at time zero and release it in the terminal year; others add increments year by year as sales grow; a few do not recover it at all. Follow the question, say so in one line, and check the Study Material for your attempt.*

FORMULA

Internal Rate of Return and Modified IRR

FORMULA

IRR = the rate at which NPV = 0 By interpolation: IRR = L + [NPV at L / (NPV at L - NPV at H)] x (H - L) MIRR: Terminal Value = Sum of [Cash inflow of year t x (1 + k)^(n - t)] MIRR = (Terminal Value / Initial outlay)^(1/n) - 1

TERMS

L = lower rate (positive NPV), H = higher rate (negative NPV), n = life, k = reinvestment

rate.

WATCH OUT

The interpolation denominator is the **difference** between the two NPVs, which becomes a sum in arithmetic when NPV at H is negative. Interpolation is an **approximation** — a straight line drawn through a curved NPV profile — so the error shrinks as L and H close in. Bracket the answer within a few percentage points and call the figure approximate.

WORKED PROBLEM

One project, every appraisal method

GIVEN

Outlay ₹1,00,000 now. After-tax inflows ₹40,000, ₹40,000, ₹30,000, ₹30,000. Life 4 years, straight-line depreciation, no salvage. Cost of capital 10%. Use the four-figure PV factors exactly as printed: at 10%, 0.9091, 0.8264, 0.7513, 0.6830; at 15%, 0.8696, 0.7561, 0.6575, 0.5718; at 18%, 0.8475, 0.7182, 0.6086, 0.5158. Round each product to the rupee.

ATTEMPT IT

Compute payback, discounted payback, ARR on both bases, NPV, profitability index, IRR by interpolation between 15% and 18%, and MIRR.

THE SOLUTION

Compute payback, discounted payback, ARR on both bases, NPV, profitability index, IRR by interpolation between 15% and 18%, and MIRR.

SOLUTION

Yr	Cash inflow (₹)	PV factor @10%	Present value (₹)	Cumulative cash (₹)	Cumulative PV (₹)	Profit after depreciation (₹)
1	40,000	0.9091	36,364	40,000	36,364	15,000
2	40,000	0.8264	33,056	80,000	69,420	15,000
3	30,000	0.7513	22,539	1,10,000	91,959	5,000
4	30,000	0.6830	20,490	1,40,000	1,12,449	5,000
			1,12,449			40,000

Depreciation = 1,00,000/4 = ₹25,000 a year, hence the profit column. Every present value here is the inflow times the **printed** factor, so the table can be reproduced digit for digit; discounting at an unrounded 10% would give ₹1,12,451, and a ₹2 difference of that kind is a

rounding convention, not an error — say which you used.

Payback — ₹80,000 is recovered by the end of Year 2, leaving ₹20,000 against Year 3's ₹30,000: 2 + 20,000/30,000 = **2.67 years**, i.e. 2 years 8 months.

Discounted payback — ₹91,959 recovered by Year 3, leaving ₹8,041 against Year 4's ₹20,490: 3 + 8,041/20,490 = **3.39 years**.

ARR — average profit = 40,000/4 = ₹10,000. On initial investment: 10,000/1,00,000 = **10%**. On average investment of (1,00,000 + 0)/2 = ₹50,000: 10,000/50,000 = **20%**.

NPV = 1,12,449 − 1,00,000 = **₹12,449**. **PI** = 1,12,449/1,00,000 = **1.1245** (which is 1 + 12,449/1,00,000, as it must be).

IRR. At 15% the present values are 34,784 + 30,244 + 19,725 + 17,154 = ₹1,01,907, so NPV = **+ ₹1,907**. At 18% they are 33,900 + 28,728 + 18,258 + 15,474 = ₹96,360, so NPV = **−₹3,640**. Interpolating:

IRR = 15 + [1,907 / (1,907 − (−3,640))] x (18 − 15) = 15 + (1,907 / 5,547) x 3 = 15 + 0.3438 x 3 = 15 + 1.03 = **16.03%**

The true IRR is 15.9988%, so the interpolation is 0.03 percentage points high — acceptable, and it would have been worse had the bracket been 10% and 20%.

MIRR — compound each inflow forward to Year 4 at 10%: 40,000 x 1.331 = 53,240; 40,000 x 1.21 = 48,400; 30,000 x 1.10 = 33,000; 30,000 x 1 = 30,000. Terminal value = **₹1,64,640**. MIRR = (1,64,640/1,00,000)^(1/4) − 1 = 1.6464^0.25 − 1 = **13.27%**.

CHECK

Four checks. NPV is positive, so IRR must exceed the 10% cost of capital and PI must exceed 1 — both hold. Discounted payback (3.39 years) must exceed plain payback (2.67), discounting shrinking every inflow. ARR on average investment must be exactly twice ARR on initial investment when there is no salvage — 20% and 10%, so it is. And MIRR (13.27%) must sit **between** the cost of capital and the IRR, because it replaces reinvestment at 16% with reinvestment at 10%; a MIRR above the IRR means you compounded at the wrong rate.

Decision rules, and where they conflict. Accept if NPV > 0; IRR > cost of capital; PI > 1; payback within the cut-off; ARR above target. For one conventional project the discounted methods agree. They diverge on **ranking**, and that is what is tested:

CONFLICT	WHY IT ARISES	WHICH TO PREFER
NPV against IRR	Different project sizes, or timing patterns; IRR assumes reinvestment at itself	NPV — it maximises absolute wealth
NPV against PI	PI favours small, capital-efficient projects	NPV, unless capital is rationed
IRR against MIRR	IRR reinvests at IRR, MIRR at the cost of capital	MIRR — the reinvestment assumption is realistic
Any of them against payback or ARR	Payback ignores post-cut-off flows; ARR ignores time value	The discounted method

Multiple IRRs arise on non-conventional flows — more than one change of sign, as with a mid-life overhaul or a terminal restoration cost — and there can be as many IRRs as sign changes. On flows of −₹1,000, +₹3,000, −₹2,100 the NPV equation is satisfied at **11.27%** and again at **88.73%**, and neither is "the" return. A decisive argument for NPV or MIRR, which give one answer.

Capital rationing ranks projects when capital is limited. Where projects are **divisible**, rank by profitability index and work down the list. Where they are **indivisible**, PI ranking can fail, and you must examine feasible combinations:

Project	Outlay (₹)	NPV (₹)	PI	PI rank
A	50,000	16,000	1.320	4
B	40,000	14,000	1.350	1
C	30,000	10,000	1.333	2
D	20,000	6,500	1.325	3

Capital available ₹1,00,000
Greedy by PI rank: B + C + D = ₹90,000 outlay, total NPV ₹30,500 (₹10,000 idle)
Best combination: A + C + D = ₹1,00,000 outlay, total NPV ₹32,500

Equivalent Annual Cost (EAC) compares assets of **unequal life** by turning each one's present value of cost into a level annual charge: EAC = PV of costs / PVAF for that life. Machine X — ₹1,00,000 plus ₹20,000 a year for 3 years at 10%: PV = 1,00,000 + 20,000 x 2.4869 = ₹1,49,738, so EAC = 1,49,738/2.4869 = **₹60,211**. Machine Y — ₹1,50,000 plus ₹15,000 a year for 5 years: PV = 1,50,000 +

15,000 x 3.7908 = ₹2,06,862, so EAC = 2,06,862/3.7908 = **₹54,569**. Choose Y: lower annual cost despite the larger total PV, because it serves longer.

EXAM TRAP

Comparing total present values of two assets with different lives

Machine Y above has the higher PV of cost (₹2,06,862 against ₹1,49,738) and is still the better buy. Comparing raw PVs silently assumes the two assets do the same job for the same period. Convert to an annual figure — EAC for costs, equivalent annual benefit for inflows — or extend both to a common horizon. The same warning applies to NPV when lives differ.

Dividend decisions

A dividend distributes earnings to shareholders, so the decision is really about how much to **retain** — every rupee paid out is a rupee not reinvested. Keep the four instruments apart:

	WHAT HAPPENS	EFFECT
Cash dividend	Cash leaves the company	Reserves and cash fall; number of shares unchanged
Stock dividend (bonus issue)	Reserves are capitalised into new fully paid shares	No cash moves; shares rise, reserves fall, EPS and price per share fall proportionately, total wealth unchanged
Stock split	The face value of each share is subdivided	Shares rise, capital unchanged, reserves unchanged; price falls proportionately; improves marketability
Share buy-back	The company purchases its own shares	Cash and shares both fall; EPS rises on the reduced base; a way of returning surplus cash and signalling confidence

Determinants of dividend: level and stability of earnings; liquidity, since profit is not cash; access to and cost of external finance; investment opportunities and their return relative to Ke; shareholders' expectations and the shareholder base; the desire for a stable payout;

contractual and statutory restrictions; signalling; and shareholders' tax position.

FORMULA

Walter's model and Gordon's model

FORMULA

Walter: P = [D + (r/Ke)(E - D)] / Ke Gordon: P0 = E1 x (1 - b) / (Ke - b x r) where growth g = b x r

TERMS

P = price per share, D = dividend per share, E = earnings per share, r = return on retained earnings, Ke = cost of equity, b = retention ratio.

WATCH OUT

Both models make payout **relevant** and both reach the same conclusion. When r > Ke (growth firm) price is highest at **zero** payout; when r < Ke (declining firm) at **100%** payout; when r = Ke payout is irrelevant. Gordon's formula breaks down once b x r reaches Ke, the denominator turning zero or negative — say so rather than reporting an absurd price.

REMEMBER THIS

With E = ₹10, r = 15%, Ke = 12%, Walter gives ₹104.17 at zero payout, ₹91.67 at a ₹6 dividend and ₹83.33 at full payout. Reverse r and Ke and the ordering reverses exactly.

Check the middle figure: [6 + (0.15/0.12)(10 − 6)]/0.12 = 11/0.12 = **₹91.67**. Gordon on the same firm at b = 40%: g = 6%, P0 = 6/(0.12 − 0.06) = **₹100**; at b = 60%: g = 9%, P0 = 4/0.03 = **₹133.33** — higher retention, higher price, because r beats Ke. Note that Walter's E is the **current** EPS while Gordon's E1 is **next year's**; a question giving a single earnings figure means both, as here, and if it gives E0 and a growth rate, take E1 = E0(1 + g) for Gordon.

The traditional approach (Graham and Dodd) also treats dividends as relevant, weighting the dividend more heavily than retained earnings in the market price. **MM's hypothesis** says the opposite: in a perfect market with no taxes or transaction costs and a given investment policy, dividend policy is **irrelevant** — a shareholder wanting cash can sell shares, one who does

not can reinvest. Pay a dividend and the firm must raise the same sum elsewhere, the fall in terminal price exactly offsetting the dividend.

PRACTISE IT

A company has 10,000 shares at ₹100, Ke 10%, expected earnings ₹1,00,000 and planned investment ₹1,50,000. Prove MM's irrelevance by computing the value of the firm when it pays a ₹10 dividend and when it pays nothing.

MODEL ANSWER

Compare against what a full-mark answer contains

Use the two MM relations: the year-end price **P1 = P0(1 + Ke) − D1**, and the funds to be raised by a fresh issue = Investment − (Earnings − Total dividend), so the number of new shares is that amount divided by P1. Then value the existing shares as

n x P0 = [(n + Δn) x P1 − (Investment − Earnings)] / (1 + Ke)

Case 1, dividend ₹10 per share. P1 = 100(1.10) − 10 = ₹100. Dividend paid = ₹1,00,000, so retained earnings are nil and the whole ₹1,50,000 must be raised: Δn = 1,50,000/100 = 1,500 shares. n x P0 = [(10,000 + 1,500) x 100 − (1,50,000 − 1,00,000)]/1.10 = (11,50,000 − 50,000)/1.10 = **₹10,00,000**.

Case 2, no dividend. P1 = 100(1.10) − 0 = ₹110. Earnings of ₹1,00,000 are retained, so only ₹50,000 need be raised: Δn = 50,000/110 = 454.55 shares. n x P0 = [(10,000 + 454.55) x 110 − 50,000]/1.10 = (11,50,000 − 50,000)/1.10 = **₹10,00,000**.

Identical, and the identity is the answer: paying the dividend transferred ₹1,00,000 of value from the terminal share price into the shareholders' pockets and forced the company to sell 1,500 shares to strangers instead of 455. Total wealth did not move.

Close by naming the assumptions, where the remaining marks are: perfect capital markets with costless information, no taxes, no transaction or floatation costs, a fixed investment policy, and certainty of future earnings. Relax any one — taxes and issue costs especially — and dividend policy becomes relevant again, which is why Walter and Gordon exist.

QUICK RECALL

A company announces a bonus issue and its share price halves the next day. Have shareholders lost money?

No. A bonus issue capitalises reserves into shares: the share count rises, the price per share falls proportionately, each holder's total value is unchanged, and no cash has left the company — which is why a bonus is no substitute for a cash dividend, and why a stock split produces the same

price effect by a different route (face value, not reserves).

```
UNIT 3 FORMULA SHEET — investment appraisal and dividend models
  k = cost of capital    n = life in years    t = tax rate    PVAF = present value annuity factor

APPRAISAL
  Payback period            = Years fully recovered + (Unrecovered cost / Next year's cash flow)
                            = Initial outlay / Annual inflow      (only if inflows are even)
  Discounted payback        = as above, using discounted cash flows
  ARR                       = (Average annual profit after tax / Investment) x 100
     Average investment     = (Initial outlay + Salvage value) / 2      -- state which base you use
  NPV                       = Sum [ CFt / (1 + k)^t ] - Initial outlay
  Profitability Index       = PV of inflows / Initial outlay = 1 + (NPV / Outlay)
  IRR                       = rate at which NPV = 0
     by interpolation       = L + [ NPV(L) / (NPV(L) - NPV(H)) ] x (H - L)      -- approximate
  MIRR                      = (Terminal Value / Outlay)^(1/n) - 1
     Terminal Value         = Sum [ CFt x (1 + k)^(n - t) ]
  Equivalent Annual Cost    = PV of all costs / PVAF(k, n)
  Annual cash flow          = (Profit before depreciation and tax) x (1 - t) + (Depreciation x t)
  Working capital           = outflow when committed, NOT depreciated, recovered untaxed at the end

  Accept if: NPV > 0 | IRR > k | PI > 1 | payback within cut-off | ARR > target
  On ranking conflicts prefer NPV; prefer PI only when capital is rationed and projects divisible.
  Non-conventional flows (more than one sign change) can give MULTIPLE IRRs - use NPV or MIRR.

DIVIDEND MODELS
  Walter          P  = [ D + (r/Ke)(E - D) ] / Ke
  Gordon          P0 = E1(1 - b) / (Ke - br)          growth g = b x r ; needs br < Ke
     r > Ke -> pay nothing | r < Ke -> pay everything | r = Ke -> payout is irrelevant
  MM              P0 = (D1 + P1) / (1 + Ke)      so  P1 = P0(1 + Ke) - D1
     New funds to raise = Investment - (Earnings - Total dividend)
     New shares  Dn     = New funds / P1
     n x P0 = [ (n + Dn) x P1 - (Investment - Earnings) ] / (1 + Ke)      -- same under any payout
```

Unit 4 — Management of Working Capital

PRINCIPAL

> The client is profitable, has a full order book, and could not pay salaries on the 7th. How?

YOU

> Their debtors are slow.

PRINCIPAL

> Debtors are slow, stock is high and the suppliers have stopped waiting. Put a number on it.

MARGIN NOTE

> *The number is the cash conversion cycle. This unit is examined as arithmetic, not opinion.*

Gross working capital is total current assets; **net working capital** is current assets less current liabilities. Split it another way and the financing question appears: **permanent (core) working capital** is the minimum level always tied up, and **temporary (fluctuating) working capital** rises and falls with the season or the order book.

Liquidity against profitability is the trade-off the unit turns on. Holding more current assets, and financing them long term, makes the firm safe and less profitable: idle cash earns little, generous credit buys sales at a cost, large stocks avoid stock-outs and tie up funds. Holding less does the opposite. Three financing strategies name the positions:

STRATEGY	HOW IT FINANCES	EFFECT
Conservative	Long-term funds cover permanent needs and part of the fluctuating need	Low risk, low return; surplus funds in slack season
Aggressive	Short-term funds cover the fluctuating need and part of the permanent need	High return, high refinancing and interest-rate risk
Moderate (matching or hedging)	Long-term funds match permanent needs, short-term funds match fluctuating needs	Maturity of the finance matches the life of the asset

Primary and secondary sources of liquidity must be kept apart. **Primary sources** are the ordinary, low-cost,

day-to-day ones: cash and near-cash balances, cash generated by trading, trade credit taken, and undrawn bank limits. **Secondary sources** are reached for when the primary ones are exhausted: renegotiating or rescheduling debt, factoring or securitising receivables, liquidating inventory at a discount, selling long-term assets, and ultimately insolvency proceedings. The distinction is not merely descriptive — **using a secondary source changes the firm's operations or capital structure and signals distress**; using a primary source does not.

The operating cycle and the cash conversion cycle

FORMULA

The operating cycle, the cash conversion cycle and the funds they need

FORMULA

Raw material period = (Average raw material stock / Annual RM consumption) x 365 Work-in-progress period = (Average WIP / Annual cost of production) x 365 Finished goods period = (Average finished goods / Annual

cost of goods sold) x 365 Receivables period = (Average debtors / Annual credit sales) x 365 Payables period = (Average creditors / Annual credit purchases) x 365

Gross operating cycle = RM period + WIP period + FG period + Receivables period Net operating cycle (cash conversion cycle) = Gross operating cycle - Payables period Number of operating cycles a year = 365 / Net operating cycle Working capital required = Total operating cost / Number of operating cycles a year

TERMS

Each numerator is a stock (a balance), each denominator a flow (a year's turnover). Use averages where opening and closing figures are given, closing balances otherwise.

WATCH OUT

Every ratio must pair a stock with **its own** flow — finished goods against cost of goods sold, not sales; debtors against **credit** sales; creditors against **credit purchases**, not consumption. Pairing debtors with cost of sales, or stock with sales, loses the whole answer.

Verify: *conventions differ. Many texts define the operating cycle as inventory period plus receivables period and reserve "cash conversion cycle" for that figure minus payables — the same arithmetic under different names, as above. Some questions use 360 days rather than 365. Follow the question's basis, state it, and check the Study Material for your attempt.*

PRACTISE IT

Average raw material ₹40,000 against annual consumption ₹3,00,000; average WIP ₹30,000 against cost of production ₹4,50,000; average finished goods ₹50,000 against cost of goods sold ₹5,00,000; average debtors ₹80,000 against credit sales ₹6,00,000; average creditors ₹45,000 against credit purchases ₹3,20,000. Total operating cost ₹5,50,000. Compute the operating cycle, the cash conversion cycle and the working capital requirement.

MODEL ANSWER

Compare against what a full-mark answer contains

Take a 365-day year and pair each stock with its own flow.

```
Raw material period   (40,000 / 3,00,000) x 365   =   48.67 days
Work-in-progress      (30,000 / 4,50,000) x 365   =   24.33 days
Finished goods        (50,000 / 5,00,000) x 365   =   36.50 days
Receivables           (80,000 / 6,00,000) x 365   =   48.67 days
                                                      -----------
GROSS OPERATING CYCLE                               158.17 days
Less: Payables period (45,000 / 3,20,000) x 365   =   51.33 days
                                                      -----------
NET OPERATING CYCLE (CASH CONVERSION CYCLE)         106.84 days
```

The net cycle is 106.8385 days unrounded, so operating cycles a year = 365 / 106.8385 = **3.4164 cycles**.

Working capital required = 5,50,000 / 3.4164 = **₹1,60,990**, call it ₹1.61 lakh.

Cross-check the other way, since either may be asked: 5,50,000 x 106.8385/365 = **₹1,60,990**. The two routes are algebraically the same statement, so if they disagree you have made an arithmetic slip, not a methodological choice.

Carry the unrounded cycle. Rounding the cycle count to 3.42 gives ₹1,60,819 —

about **₹171 out**. Rounding the *days* to 106.84 first gives ₹1,60,992, which is ₹2 out. Neither error is large enough to matter commercially and both are large enough to lose a tick, so keep full precision in the working and round only the final answer. If you must show a rounded cycle figure, show it and then say you divided by the unrounded one.

The sentence that earns the interpretation mark: cash is locked up for roughly 107 days, so a **shorter** cycle releases cash with no new finance — collect faster, hold less stock, pay later — and each day removed cuts the funds needed by about ₹1,507 (5,50,000/365).

Assessment of the working capital requirement

Three approaches are examined. The **estimation of components** method builds current assets and liabilities up line by line, valuing stock and debtors at cost and adding a stated safety margin — noting that debtors may be taken at **cost** or at **selling price** depending on the question, and that cash cost excludes depreciation. The **percentage of sales** method applies a historical ratio of working capital to sales: quick and crude. The **operating cycle** method is the one just worked. Whichever is used, state whether you worked on a **total cost** or a **cash cost** basis.

Receivables, payables and credit management

Credit granting assesses the applicant — character, capacity, capital, collateral, conditions — then sets the credit period, any early-payment discount and the credit limit. **Monitoring** uses the average collection period, an **ageing schedule** of receivables by days outstanding, and receivables as a proportion of sales. **Collection** escalates from statements and reminders to telephone follow-up, withdrawal of credit, an agency, and legal action. The

governing rule: **relax credit only while the incremental contribution exceeds the incremental cost of bad debts, discounts, administration and the funds tied up.**

FORMULA

The cost of forgoing a cash discount

FORMULA

Cost (% per annum) = [d / (100 - d)] x [365 / (Credit period - Discount period)] x 100

TERMS

d = discount percentage offered for early payment.

WATCH OUT

This is the implicit annual interest in *not* taking a discount, usually far above any borrowing rate — on 2/10, net 45 it is (2/98) x (365/35) = **21.28%** a year, so a firm that can borrow more cheaply should always take the discount. Read the same formula the other way to price the discount you offer your own customers.

Managing payables is the mirror image: trade credit is free while no discount is forgone, so take the full period

offered — but stretching beyond it costs goodwill, invites supply interruption and eventually cash-on-delivery terms. Compare the cost of forgoing the discount with the firm's own cost of short-term finance and choose the cheaper.

Factoring and forfaiting are both ways of turning receivables into cash, and the comparison is examinable almost every attempt:

	FACTORING	FORFAITING
Receivables covered	Ordinary trade receivables, domestic or export	Export receivables only
Instrument	Invoices; no negotiable instrument needed	Bills of exchange or promissory notes, usually bank-guaranteed
Tenor	Short-term	Medium to long-term
Recourse	With or without recourse	Always without recourse

	FACTORING	FORFAITING
Amount advanced	Part of invoice value, the balance on collection	Effectively the whole value, discounted upfront
Services	Finance plus sales ledger administration, collection and credit protection	Financing only

WORKED PROBLEM

Should the credit period be extended from 30 days to 60?

GIVEN

Present sales ₹10,00,000, 30-day collection period, bad debts 1% of sales. Variable cost 70% of sales; fixed cost ₹1,50,000. Extending credit to 60 days raises sales to ₹12,00,000 with bad debts of 2%; fixed cost unchanged. Required return on funds in receivables 20% a year. Use 365 days and value debtors at total cost.

ATTEMPT IT

Evaluate the proposal, showing incremental contribution, incremental bad debts and the opportunity cost of the extra investment in receivables.

THE SOLUTION

Evaluate the proposal, showing incremental contribution, incremental bad debts and the opportunity cost of the extra investment in receivables.

SOLUTION

	Present (30 d)	Proposed (60 d)	Incremental
Sales	10,00,000	12,00,000	
Less: Variable cost at 70% of sales	7,00,000	8,40,000	
CONTRIBUTION	3,00,000	3,60,000	60,000
Cost build-up for valuing debtors:			
Variable cost at 70% of sales	7,00,000	8,40,000	
Add: Fixed cost	1,50,000	1,50,000	
TOTAL COST	8,50,000	9,90,000	
Debtors at total cost x days/365	69,863	1,62,740	92,877
Bad debts (1% / 2% of sales)	10,000	24,000	14,000

Opportunity cost of the extra investment = 92,877 x 20% = **₹18,575.**

Net benefit = incremental contribution 60,000 − incremental bad debts 14,000 − opportunity

cost 18,575 = **₹27,425.** The proposal is worth accepting.

CHECK

Check each term's direction before trusting the total. Fixed cost appears in both columns and cancels out of the *incremental* contribution, which is why 30% of the ₹2,00,000 extra sales (₹60,000) is the right figure. Bad debts rise on both counts — bigger base, worse rate. And the investment in debtors roughly doubles because both the cost base and the period rise; a small incremental investment means you applied 30/365 rather than 60/365 to the proposed column.

EXAM TRAP

Valuing debtors at sales value when the question says cost

Money is tied up in debtors only to the extent of the **cost** incurred — the profit element was never spent, so no funds are locked in it. ICAI questions therefore say "total cost" or "cash cost", and the opportunity cost is sensitive to

which: debtors at sales value here would be 12,00,000 x 60/365 = ₹1,97,260 instead of ₹1,62,740. State your basis in one line, and remember cash cost excludes depreciation.

Cash and treasury management

Three motives for holding cash: **transaction** (routine payments), **precautionary** (a buffer against uncertainty) and **speculative** (to exploit an opportunity). The control tool is the **cash budget**, projecting receipts and payments period by period and exposing deficits before they arrive. Cash is then managed by **speeding collections** (prompt billing, decentralised collection, concentration banking, lock boxes) and **controlling disbursements** (centralised payment, use of the cheque float, paying on the last permissible day), with genuine surpluses invested short term.

Treasury management owns all of this: liquidity and cash-flow management, banking relationships and the negotiation of facilities, arranging short-term funding, investing surpluses, monitoring the firm's financial exposures and reporting on them. Its aim is the right money in the right place at the right time — no idle surplus, no missed payment.

> ***Verify:*** *the Study Material develops optimum cash balance models — associated with Baumol and with Miller and Orr — under management of cash. Whether they are examinable, and in what depth, is set by the Study Guidelines for your attempt; confirm before spending revision time on them.*

Banking norms of working capital finance

A bank does not finance the whole working capital gap. The borrower contributes a **margin** from its own long-term funds and the bank funds the balance. Against the security charged — usually stocks and book debts — the bank computes a **drawing power**: eligible security value, less the stipulated margin, less unpaid creditors for those stocks. Drawings are allowed up to the **lower** of drawing power and the sanctioned limit, which is why a borrower with an adequate limit can still be unable to draw. The underlying idea is a **maximum permissible bank finance** — a ceiling derived from the working capital gap after the borrower's own contribution — policed by periodic stock and book-debt statements and an annual review.

Verify: *the specific margin percentages, the method by which permissible bank finance is computed, and the eligibility rules for security are matters of banking regulation and lender policy, and they change. This note deliberately states no percentage. Confirm the treatment expected in the Study Material and Study Guidelines for your attempt.*

REMEMBER THIS

A shorter cash conversion cycle is free finance. Every day removed releases funds permanently and costs no interest, which is why the cycle is the first thing to compute and the last thing to report.

On the figures above one day is about ₹1,507, so cutting the 48.67-day receivables period to 40 days releases roughly ₹13,000 of the ₹1,60,992 requirement with no new borrowing. Benefit against the cost of achieving it is the form every working capital answer should take.

QUICK RECALL

Current ratio is 2.5 and rising, and the cash conversion cycle is lengthening. Is liquidity improving?

No — it is deteriorating while the ratio flatters it. A current ratio rising on slower stock and slower-paying debtors means more funds locked up for longer, not more cash available. The ratio is a snapshot of balances; the cycle measures how fast they turn into cash. Read them together.

REVISION CHECKLIST

Formulas to reproduce without notes. The complete Section A set, in one block:

```
UNIT 1 — RATIOS
  Current ratio = CA/CL   Quick = (CA - Inventory)/CL
  Debt-equity = Long-term debt / Shareholders' funds
  Interest coverage = EBIT / Interest
  DSCR = (PAT + Depreciation + Interest on term loan) / (Interest + Principal instalment)
  Inventory turnover = COGS / Average inventory        Days = 365 / turnover
  Debtor turnover = Credit sales / Average debtors   Creditor days = (Creditors/Purchases) x 365
  ROCE = EBIT / Capital employed        Capital employed = Total assets - Current liabilities
  ROE = PAT / Shareholders' funds = Net profit margin x Asset turnover x Equity multiplier
  EPS = [(EBIT - I)(1 - t) - Dp] / Number of equity shares

UNIT 2 — COST OF CAPITAL, STRUCTURE, LEVERAGE
  Ke = D1/P0 + g  (or D1/NP + g)      Ke = Rf + Beta(Rm - Rf)      g = b x r
  Kp = PD/NP ;  redeemable = [PD + (RV - NP)/n] / [(RV + NP)/2]     -- never x (1 - t)
  Kd = I(1 - t)/NP ;  redeemable = [I(1 - t) + (RV - NP)/n] / [(RV + NP)/2]
  WACC = Sum(weight x cost), weights on book OR market values; MCC = cost of the NEW funds only
  V = EBIT/Ko    Ke = (EBIT - I)/E    MM with tax: Vl = Vu + Dt
  Indifference EBIT: [(E - I1)(1-t) - Dp1]/n1 = [(E - I2)(1-t) - Dp2]/n2
  Financial break-even EBIT = I + Dp/(1 - t)
  DOL = Contribution/EBIT    DFL = EBIT/PBT    DFL(pref) = EBIT/[EBIT - I - Dp/(1-t)]
  DCL = DOL x DFL = %dEPS/%dSales              -- a product, never a sum
        Contribution/PBT ONLY if Dp = 0; else Contribution/[EBIT - I - Dp/(1-t)]

UNIT 3 — APPRAISAL AND DIVIDEND
  Payback = Years recovered + Unrecovered cost / Next year's flow
  ARR = Average annual profit after tax / Investment      Avg investment = (Outlay + Salvage)/2
  NPV = Sum[CFt/(1+k)^t] - Outlay        PI = PV of inflows / Outlay = 1 + NPV/Outlay
  IRR = L + [NPV(L)/(NPV(L) - NPV(H))] x (H - L)           -- an approximation
  MIRR = (Terminal Value / Outlay)^(1/n) - 1    TV = Sum[CFt x (1+k)^(n-t)]
  EAC = PV of costs / PVAF(k, n)
  Walter: P = [D + (r/Ke)(E - D)]/Ke      Gordon: P0 = E1(1-b)/(Ke - br)
  MM: P1 = P0(1 + Ke) - D1 ;  New shares = [Investment - (Earnings - Dividend)] / P1

UNIT 4 — WORKING CAPITAL
  Each period = (Average balance / its own annual flow) x 365
  Gross operating cycle = RM + WIP + FG + Receivables periods
  Cash conversion cycle = Gross operating cycle - Payables period
  Working capital = Total operating cost / (365 / cash conversion cycle)
  Cost of forgoing a discount = [d/(100-d)] x [365/(credit - discount period)] x 100
```

Decision rules, and where they fight each other.

- Accept a project if **NPV > 0**, **IRR > cost of capital**, **PI > 1**, payback is within the cut-off, or ARR beats the target. For one conventional project the discounted measures agree.
- On **ranking**, prefer **NPV** — it maximises absolute wealth. Prefer **PI** only when capital

is rationed and projects are divisible; with indivisible projects, test feasible combinations, because PI ranking can leave funds idle and money on the table.

- Prefer **MIRR to IRR** where the reinvestment assumption matters, and abandon IRR altogether when cash flows change sign more than once, because **multiple IRRs** exist.
- Never rank on **payback** (ignores everything after the cut-off and, undiscounted, the time value of money) or **ARR** (uses profit, not cash, and ignores timing) — quote them as supplementary.
- In financing, accept the geared plan only **above the indifference EBIT**; in dividends, follow Walter and Gordon's single rule — retain when r > Ke, distribute when r < Ke.

Ratios and what each one diagnoses. Liquidity — current, quick, cash: can short-term obligations be met. Solvency — debt–equity, proprietary, capital gearing, interest coverage, DSCR: how much risk sits with lenders and whether they can be serviced. Activity — inventory, receivables, payables, fixed asset,

working capital turnover: how hard the assets are worked and how long cash is locked up. Profitability — gross, operating and net profit ratios, ROCE, ROE, return on assets: whether the trading and the capital both pay. Market — EPS, price–earnings, earnings and dividend yield, payout: what the market will pay for the earnings. Remember that ROCE is the referee between an efficiency story and a gearing story, and that every ratio is a symptom, not a cause.

Traps that cost marks. Multiplying Ke or Kp by (1 − t). Adding leverages instead of multiplying them. Using EBIT/PBT for DFL when preference shares exist. Giving retained earnings a separate market value. Comparing raw present values of assets with unequal lives instead of EAC. Valuing debtors at sales value when the question says cost. Pairing a stock with the wrong flow in a cycle computation. Treating a bonus issue or a stock split as a gain to shareholders.

What to verify against the Study Guidelines for your attempt. Whether ARR uses average or initial investment. The working capital convention in NPV — time-zero injection against year-by-year increments, and whether

it is recovered. Whether any tax on dividend distribution enters Kp or Ke, and whether retained earnings are adjusted for personal tax and brokerage. Whether 360 or 365 days is expected, and whether averages or closing balances are used in ratios and cycles. Whether the optimum cash balance models are in scope. And the margin, drawing power and permissible bank finance rules, none of which is stated as a percentage in these notes because they are set by regulation and lender policy. These notes revise the syllabus; they do not replace the Study Material or the Guidelines.